



MAYOOR SCHOOL, NOIDA

CLASS X (2025-26)

HOLIDAY ENGAGEMENT PROGRAMME

Dear Parents

The holiday homework is meticulously designed to enrich students' learning, reinforce essential skills, and stimulate intellectual curiosity. This engaging academic endeavor ensures continuity in education, fosters self-discipline, and promotes independent thinking, providing a well-rounded, enriching experience during the break.

Warm regards

Principal's Office

Dear Students

We are excited to announce the commencement of the summer vacation, an ideal time for rejuvenation and enrichment. To maximize this period, we have crafted the holiday homework to harness your creative energies. These assignments aim to maintain your syllabus connection, enhance creativity, and engage you constructively, thereby unveiling your inherent potential. We encourage you to complete these tasks with neat, presentable and original work, adhering to the provided guidelines.

Best wishes

Principal's Office

ENGLISH

1. English Through Art: Story Collage

Unleash your artistic flair by crafting a vibrant visual collage or scrapbook page titled **“Echoes of Literature.”**

Inclusions:

Create **two comic strips** illustrating your favourite incident from the following:

- **Mrs. Packletide’s Tiger (Roll No. 1-15)**
- **The Frog and the Nightingale (Roll No. 16-35)**

- Add witty **memes or caricatures** of key characters
- Insert **Post-it style quotes** highlighting memorable lines
- Feature **a song or poem** that resonates with the story’s theme
- Include a creative twist: *“If this story took place in 2025...”*

Guidelines:

Utilize materials like newspaper clippings, maps, illustrations, and web content. Every illustration must be paired with a relevant write-up. The project should be handwritten. Maintain the following sequence on A4 sheets:

- Cover Page (Title, Name, Roll No.)
- Acknowledgement
- Index
- Main Content
- Conclusion

2. DIY Comic: If Shakespeare Had WhatsApp

Watch the movie Julius Caesar, after watching, reimagine scenes as lively **WhatsApp chats** between characters, packed with **emojis, GIFs, stickers** and **optional voice notes**. Inject humour while preserving the play’s core essence as mentioned in your English Reader.

3. Workbook Practice:

Thoroughly complete the following grammar units:

- Determiners
- Tenses
- Subject-Verb Agreement
- Reported Speech
- Modals

4. Main Course Book:

Complete Unit 1: Health & Medicine, Unit 2: Education, and Unit 3: Science.

Revise diligently for the upcoming BLOCK TEST.

MATHEMATICS

Project work - " Arithmetic Progressions in the Real World. "

- Look for examples of arithmetic progression in daily life, such as seating arrangements in a theater or stadium, population increase, etc., and present findings with images or diagrams
- Create your own sequences and determine if they are APs. Vary the common difference and observe how it affects the sequence.
- Choose a sequence of your choice (e.g., starting from 1 with a common difference of 3) and calculate the sum of the first 10 terms both manually and using the formula. Compare results
- Identify and analyse arithmetic progressions in everyday situations, such as the pattern of seating in a stadium, the increasing cost of a product over time, or the accumulation of savings with a fixed regular deposit
- Collect data from real-life examples of Arithmetic Progressions from your society and friends, represent them as arithmetic progressions, calculate the common difference and the n th term, and make predictions based on your findings.
- Present your findings visually using graphs, charts, tables, or diagrams.

Presentation Tips:

- Use colorful charts and graphs to make your project visually appealing.
- Include real-life examples (photos, articles) to illustrate applications of AP.
- Submit this project in the PowerPoint presentation form and submit hard copy in a project file having cover page, acknowledgment, list of content, content, conclusion.

Worksheet

1. Three bulbs red, green and yellow flash at intervals of 80 seconds, 90 seconds and 110 seconds. All three flash together at 8:00 am. At what time will the three bulbs flash altogether again?
2. A seminar is being conducted by an Educational Organisation, where the participants will be educators of different subjects. The number of participants in Hindi, English and Mathematics are 60, 84 and 108 respectively.
 - i. In each room the same number of participants are to be seated and all of them being in the same subject, hence find the maximum number participants that can accommodated in each room.
 - ii. What is the minimum number of rooms required during the event?
 - iii. Find the LCM of 60, 84 and 108.
 - iv. What is the product of HCF and LCM of 60,84 and 108?
3. It is common that Governments revise travel fares from time to time based on various factors such as inflation (a general increase in prices and fall in the purchasing value of money) on different types of vehicles like auto, Rickshaws, taxis, Radio cab etc. The auto charges in a city comprise a fixed charge together with the charge for the distance covered. Study the following situations
 - Situation 1: In city A, for a journey of 10 km, the charge paid is Rs 75 and for a journey of 15 km, the charge paid is Rs 110.
 - Situation 2: In a city B, for a journey of 8km, the charge paid is Rs91 and for a journey of 14km, the charge paid is Rs 145.



Name of the city	Distance travelled (Km)	Amount paid (Rs.)
City A	10	75
	15	110
City B	8	91
	14	145

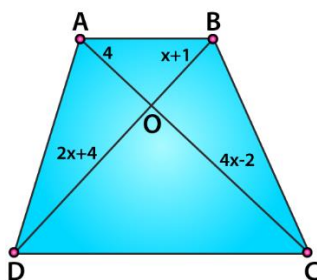
Refer situation 1

- i. If the fixed charges of an auto rickshaw are Rs x and the running charges are Rs y km/hr, then write the pair of linear equations representing the situation.
- ii. A person travels 50 km. What amount does he has to pay?
- iii. Refer situation 2: What will a person have to pay for traveling 30km?

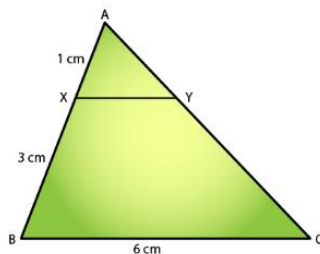
4. John and Priya went for a small picnic. After having their lunch Priya insisted on traveling in a motorboat. The speed of the motorboat was 20 km/hr. Priya being a Mathematics student wanted to know the speed of the current. So, she noted the time for upstream and downstream. She found that for covering the distance of 15 km the boat took 1 hour more for upstream than downstream.
 - i. Let the speed of the stream be x km/hr. Then what will be the speed of the motorboat upstream?
 - ii. Write the correct quadratic equation for the speed of the current?
 - iii. What is the speed of current?
5. India is a competitive manufacturing location due to the low cost of manpower and strong technical and engineering capabilities contributing to higher quality production runs. The production of TV sets in a factory increases uniformly by a fixed number every year. It produced 16000 sets in 6th year and 22600 in 9th year.
 - i. Find the production during the first year.
 - ii. Find the production during 8th year.
 - iii. Find the production during the first 3 years.
 - iv. In which year, the production is Rs 29,200?
 - v. Find the difference of the production during 7th year and 4th year.
6. If α and β are the zeroes of polynomial $3x^2 + 2x - 6$, then find the value of the following:

(i) $\alpha - \beta$ (ii) $\alpha^2 + \beta^2$ (iii) $\alpha^3 + \beta^3$ (iv) $\frac{1}{\alpha} + \frac{1}{\beta}$
7. Find the zeroes of $\sqrt{3}x^2 + 10x + 7\sqrt{3}$ and verify the relationship between the zeroes and the coefficients.
8. If the roots of quadratic equation $(b - c)x^2 + (c - a)x + (a - b) = 0$ are equal then prove that $2b = a + c$
9. If the roots of quadratic equation $(1 + m^2)x^2 + 2mcx + c^2 - a^2 = 0$ has equal roots then prove that $c^2 = a^2(1 + m^2)$
10. Prove that both the roots of $(x - a)(x - b) + (x - b)(x - c) + (x - c)(x - a) = 0$ are real but they are equal only when $a = b = c$
11. A natural number when increased by 12, equals 160 times its reciprocal. Find the number.
12. If a man had walked 2 km/hr faster, he would have taken 40 minutes less to walk 5 km. Find the speed of walking.
13. A motor boat whose speed is 20 km/hr in still water takes 1 hour more to go 21 km upstream than to return downstream to the same spot. Find the speed of the stream.
14. A two digit number is such that the product of its digits is 18. When 63 is subtracted from the numbers the digits interchange their places. Find the number.
15. The 10th and 18th terms of an A.P. are 41 and 73, respectively. Find the 26th term.

16. Find the 12th term from the end of the following arithmetic progressions.
 (i) 3, 5, 7, 9, 201 (ii) 3, 8, 13, ... , 253 (iii) 1, 4, 7, 10, ... , 88
17. Find the second term, and the n^{th} term of an A.P. whose 6th term is 12 and the 8th term is 22.
18. How many numbers of two digits are divisible by 3?
19. The 7th term of an A.P. is 32, and its 13th term is 62. Find the A.P.
20. Find the sum of the last ten terms of the A.P.: 8, 10, 12, 14, .. , 126
21. How many terms of the sequence 18, 16, 14.... should be taken so that their sum is zero.
22. The first and the last terms of an A.P. are 17 and 350, respectively. If the common difference is 9, how many terms are there, and what is their sum?
23. The sum of the first seven terms of an A.P. is 182. If its 4th and 17th terms are in a ratio 1: 5, find the A.P.
24. In a ΔABC , D and E are points on the sides AB and AC, respectively. If AB = 12 cm, AD = 8 cm, AE = 12 cm, and AC = 18 cm then show that DE $\parallel$ BC or not.
25. In fig., if $AB \parallel CD$, find the value of x .



26. Two poles of heights 6 m and 11 m stand on a plane ground. If the distance between their feet is 12 m, find the distance between their tops.
27. In fig. given, XY $\parallel$ BC. Find the length of XY.



SCIENCE

Dear students,

"In our ongoing commitment to fostering a passion for learning and cultivating a profound understanding of the intricate concepts across diverse scientific disciplines, we seek to integrate the realms of science and art. Let's unleash the potential of our students through the following interactive learning activities."

Be at your creative best, search, try and come out with something unique and interesting. The holiday homework has been designed in accordance to the latest CBSE guidelines.

I → ART INTEGRATED PROJECT

You are required to prepare an Art-Integrated Project. The activity must creatively incorporate any form of Indian Art.

The activities have been distributed roll number wise as follows:

Roll Numbers 1-10: PHYSICS

Roll Numbers 11- 20: CHEMISTRY

Roll Numbers 21 onwards: BIOLOGY

Students are to follow the above distribution for doing the activities. Following are the activities subject-wise:

PHYSICS

Choose ONE topic from your Physics syllabus and explain or revise it using ONE creative method from the list below.

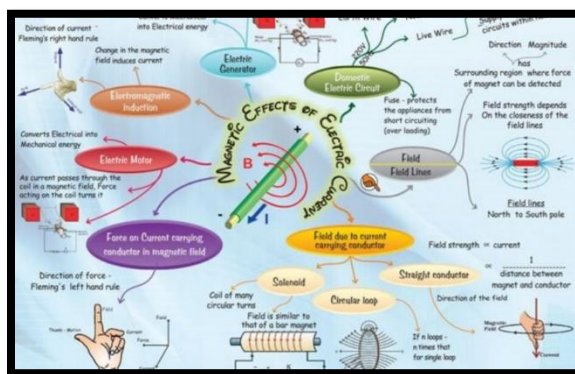
Creative Tools – Pick One!

1. □ Concept Map

Design a detailed concept map showing how key ideas in your topic are connected.

→ Use linking phrases like "*causes*", "*is related to*", "*leads to*", "*depends on*", etc.

Tip: Use colours or branches to separate subtopics.



2. 🎵 Mnemonic Magic

Invent 3–5 original mnemonics to help remember important Physics formulas, laws, or concepts.

Tip: Use rhymes, acronyms/funny phrases—they're easier to remember!

- Submit your work as a photo of your hand-drawn work, or a printout (if done digitally).
- Clearly write your Name, Class, Topic, and Creative Tool used at the top.
- Ensure content is accurate, neat, and original.

Create an infographic combining images, key terms, formulas, and quick facts.

Tip: Include “Did You Know?” facts or short definitions.

Draw or print a clear diagram of a Physics phenomenon or instrument.

Tip: Use arrows, symbols, and different colours to highlight actions or movement.



1. Art Integrated Project:

(Cross-Curricular Connection – History and Geography)

Topic: Rust-Resistant Monuments of India - A Historical and Geographical Perspective

Instructions:

Prepare a report by collecting pictures and information about historically significant rust-resistant monuments built in India. This project will explore the historical relevance of these monuments (such as their time of construction, purpose, and cultural significance)

and their geographical context (including location, climate impact, and environmental conditions that affect or preserve these structures).

Your report must include the following sections:

- **Aim:** Purpose and relevance of the project.
- **Index:** List of contents for easy navigation.
- **Content:** Detailed information and visuals about selected monuments.
- **Observation:** Key findings and insights on how history and geography influence the longevity and preservation of these structures.
- **Bibliography:** List of references and sources used.

2. Activity: Make a Short Video on Balancing Chemical Equations Using Simple Models.

Instructions:

Create a short and engaging video demonstrating the concept of balancing chemical equations. Use clay models, household items, or simple handmade models to represent atoms and molecules. Choose one example from each type of chemical equation (e.g., combination, decomposition, displacement, and double displacement reactions).

Steps to Follow:

1. Select one reaction of each type (e.g., $\text{H}_2 + \text{O}_2 \rightarrow \text{H}_2\text{O}$ for combination).
2. Use models like colored clay balls (different colors for different atoms), bottle caps, or paper cut-outs to represent atoms.
3. Show both the unbalanced and balanced forms of the equation visually using your models.
4. Explain how the number of atoms on both sides of the equation must be equal and how you achieved the balance.
5. Record your explanation clearly while showing the model demonstration.
6. Keep the video short (2–4 minutes) and creative.

Example: $\text{H}_2 + \text{O}_2 \rightarrow \text{H}_2\text{O}$

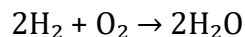
Materials:

Use 2 white balls (H) for hydrogen

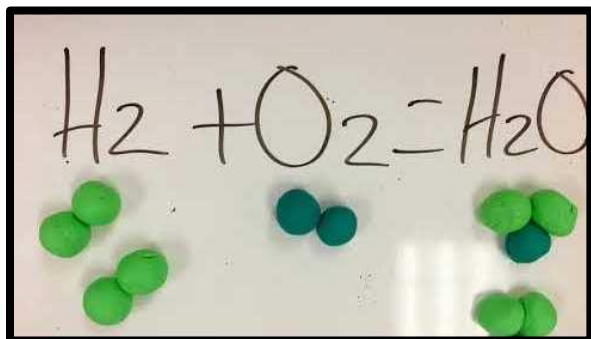
Use 2 red balls (O) for oxygen

Script:

"Here I have two white balls for hydrogen and two red balls for oxygen. In a combination reaction, substances combine to form one product. Let's balance this equation:



That means 4 hydrogen atoms and 2 oxygen atoms make 2 water molecules. Now both sides are balanced!"



You may refer to the following link for ideas and suggestion:

https://youtu.be/8ZHFZ_U6nuM

BIOLOGY

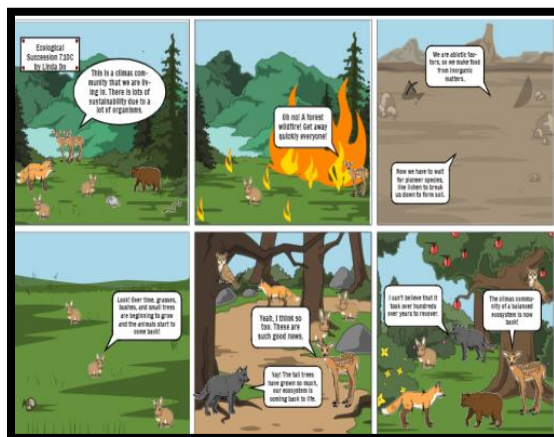
Choose ONE topic from your Biology syllabus and explain or revise it using ONE creative method from the list below.

1) Storyboard or Comic Page 🎬

Create a six-panel storyboard or comic page explaining a biological process (e.g., food digestion, blood circulation, or the nitrogen cycle).

→ Add speech bubbles, thought bubbles, and expressive characters (like talking organs or molecules).

💡 *Tip: Use humor or drama to bring the process to life!*



2) Biology Zine (Mini Booklet)

Design a small zine (DIY mini booklet) with handwritten notes, diagrams, and creative illustrations explaining your topic.

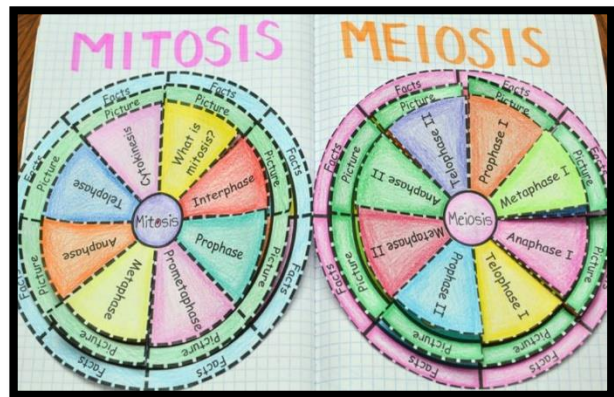
→ You can fold a single A4 page into a mini-book.

💡 *Tip: Add a "Did You Know?" fact corner or a quiz page!*

3) Foldable Infographic (Flipbook or Poster)

Create a colorful foldable or poster with flaps explaining subtopics (e.g., parts of the nephron, steps of respiration, levels in a food chain).

💡 *Tip: Use drawings, keywords, arrows, and flowcharts.*



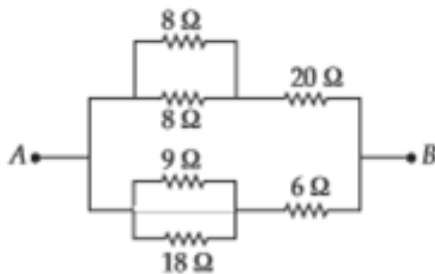
WORKSHEET

PHYSICS

Q1. Specific resistance is numerically equal to the resistance offered by

- a) 1 cm length of a conductor
- b) a conductor of unit cross-section
- c) 1 cm length of conductor of 1 cm^2 of cross-section
- d) 1 cm^3 of a conductor

Q2. The equivalent resistance between the points A and B as shown in the figure is



- a) $6\ \Omega$
- b) $8\ \Omega$
- c) $16\ \Omega$
- d) $24\ \Omega$

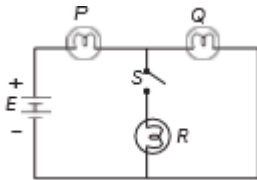
Q3. In order to distribute a high potential, we connect a number of resistors

- a) in series
- b) in parallel
- c) some in series and some in parallel
- d) It is not possible to distribute potential

Q4. Three resistors of $10\ \Omega$, $15\ \Omega$, and $20\ \Omega$, are connected in series in a circuit. If the potential drop across the 15 -ohm resistor is $3\ \text{V}$, find the potential drop across the 10 -ohm resistor.

Q5. A bulb is rated 40W , 220V . Find the current drawn by it, when it is connected to a 220V supply. If the given bulb is replaced by a bulb of rating 25W , 220V , will there be any change in the value of current and resistance?

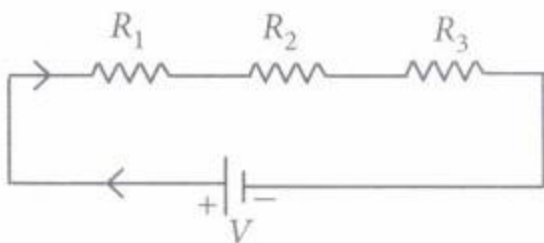
Q6. A battery E is connected to three identical lamps P , Q and R as shown in figure. Initially, switch S is kept open, and the lamp P and Q are observed to glow with same brightness. Then, switch S is closed. How will the brightness of glow of bulbs P and Q change? Justify your answer.



Q7. If you are provided with three identical resistors, in how many different ways can you combine all three resistors to obtain different effective resistances? Show all possible combinations using circuit diagrams.

Q8. Read the following paragraph and answer the given questions:

Two or more resistances are connected in series or in parallel or both, depending upon whether we want to increase or decrease the circuit resistance.



Two or more resistances are said to be connected in series if the current flowing through each resistor is same. The equivalent resistance in the series combination is given by

$$R_s = R_1 + R_2 + R_3$$

(i) When three resistors are connected in series with a battery of voltage V and voltage drop across resistors is V_1 , V_2 and V_3 , which of the relation is correct?

- (a) $V = V_1 = V_2 = V_3$ (b) $V = V_1 + V_2 + V_3$
 (c) $V_1 + V_2 + V_3 = 3V$ (d) $V > V_1 + V_2 + V_3$

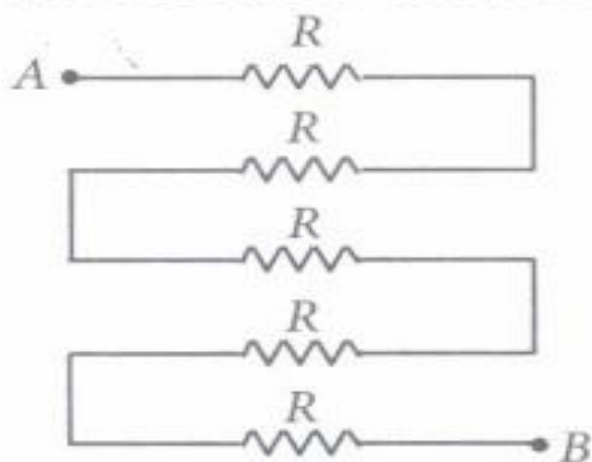
(ii) When the three resistors each of resistance R ohm, connected in series, the equivalent resistance is

- (a) $R/2$ (b) $> R$
 (c) $< R/2$ (d) $< R$

(iii) There is a wire of length 20 cm and having resistance 20Ω cut into 4 equal pieces and then joined in series. The equivalent resistance is

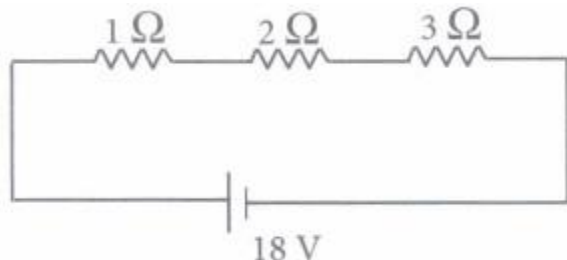
- (a) 20Ω (b) 4Ω
 (c) 5Ω (d) 10Ω

(iv) In the following circuit, find the equivalent resistance between A and B is ($R = 2 \Omega$)



- (a) $10\ \Omega$ (b) $5\ \Omega$
 (c) $2\ \Omega$ (d) $4\ \Omega$

(v) In the given circuit, the current in each resistor is



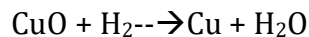
- (a) 3 A (b) 6 A
 (c) 9 A (d) 18 A

CHEMISTRY

Q1. Identify the chemical equation which represents a complete balanced equation for the reaction of barium chloride with sodium sulphate to produce barium sulphate and sodium chloride.

- a) $\text{BaCl}_2(\text{aq}) + \text{Na}_2\text{SO}_4(\text{aq}) \rightarrow \text{BaSO}_4(\text{s}) + 2\text{NaCl}(\text{aq})$
 b) $\text{BaCl}_2(\text{aq}) + \text{Na}_2\text{SO}_4(\text{aq}) \rightarrow 2\text{BaSO}_4(\text{s}) + 2\text{NaCl}(\text{aq})$
 c) $2\text{BaCl}_2(\text{aq}) + \text{Na}_2\text{SO}_4(\text{aq}) \rightarrow 2\text{BaSO}_4(\text{s}) + \text{NaCl}(\text{aq})$
 d) $\text{BaCl}_2(\text{aq}) + \text{Na}_2\text{SO}_4(\text{aq}) \rightarrow \text{BaSO}_4(\text{s}) + \text{NaCl}(\text{aq})$

Q2. When copper oxide is treated with hydrogen gas copper is produced along with water. The balanced chemical equation for the given reaction is



Which substance is oxidized in the given reaction?

- a. H_2 b. Cu c. H_2O d. CuO

Q3. What happens when dilute hydrochloric acid is added to iron fillings?

- a) Hydrogen gas and iron chloride are produced
 b) Chlorine gas and hydroxide are produced
 c) Heat is absorbed, i.e. test tube becomes cold.
 d) Iron salt and water are produced

Q4. When $\text{HCl}(\text{aq})$ is exactly neutralized by $\text{NaOH}(\text{aq})$, the hydrogen ion concentration in the resulting mixture is

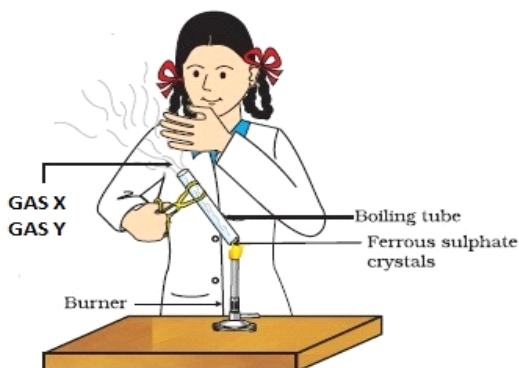
- a) always less than the concentration of the hydroxide ions
 b) always greater than the concentration of the hydroxide ions
 c) always equal to the concentration of the hydroxide ions
 d) sometimes greater and sometimes less than the concentration of the hydroxide ions

Q5. A water insoluble salt of Magnesium 'X' on reacting with dilute H_2SO_4 released a colourless and odourless gas accompanied by brisk effervescence. When the gas was passed through water, the solution obtained turned blue litmus red. On bubbling the gas through lime water, it initially became milky and milkiness disappeared when the gas was passed in excess. Name the substance 'X' and write its formula. Write the chemical equations of the reactions involved.

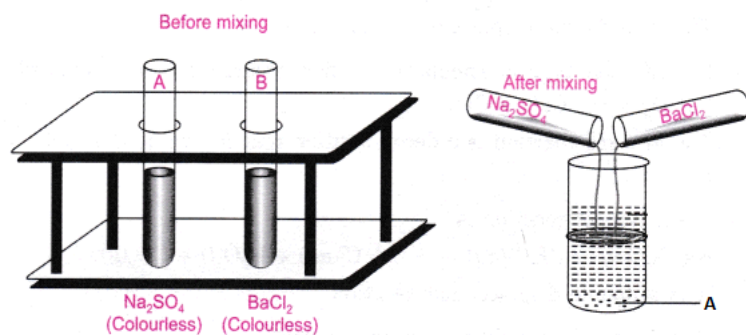
Q6. An aqueous solution of a metal nitrate 'P' reacts with sodium bromide solution to form a dull white precipitate of a compound 'Q' which is used in black and white photography. 'Q' on exposure to sunlight produces a metal and a reddish-brown gas.

- Name 'P' and 'Q' and write their formulae
- Write reactions involved in both cases.
- Identify the type of reaction in each case.

Q7. In the following figure, identify gases X and Y. Name the residue that will be left and write a balanced chemical equation of the change taking place.



Q8. In the given figure Identify substance A. Write a chemical equation for the reaction and state the type of reaction.



BIOLOGY

- Q1.** Which enzyme is responsible for the breakdown of starch in the human mouth?
(a) Lipase (b) Amylase (c) Pepsin (d) Trypsin
- Q2.** Which part of the nephron is responsible for filtration?
(a) Loop of Henle (b) Bowman's capsule (c) Collecting duct (d) Ureter
- Q3.** Which gas is primarily responsible for global warming and is also a part of the carbon cycle?
(a) Nitrogen (b) Oxygen (c) Carbon dioxide (d) Methane
- Q4.** Match the following:
a) Arteries – (i) Carry blood to heart
b) Veins – (ii) Allow exchange of gases
c) Capillaries – (iii) Carry blood away from heart
- Q5.** Why is energy flow in an ecosystem always unidirectional?
- Q6.** Draw and label the human digestive system. Name the parts where:
(a) Protein digestion begins
(b) Absorption of digested food takes place
- Q7.** Explain the importance of decomposers in the ecosystem. What would happen if there were no decomposers?
- Q8.** Read the passage and answer the questions below:

The human circulatory system includes the heart, blood, and blood vessels. The heart works as a pump and maintains the flow of blood throughout the body. Blood carries oxygen from the lungs to tissues and removes carbon dioxide. Arteries carry blood away from the heart, while veins carry it back. Capillaries connect the two and allow the exchange of gases and nutrients. If arteries become narrow due to fat deposits, it can reduce blood flow and cause heart diseases.

Answer the following:

- What type of blood does arteries usually carry?
- Name the tiny blood vessels where the exchange of gases occurs.
- What could be one possible effect of fat deposits in arteries?
- Why is the heart considered a double pump?

SOCIAL SCIENCE

1. **Crafting a Celebration of Unity: The "Ek Bharat Shreshtha Bharat" - Making of a scrapbook**

The very essence of India lies in its breathtaking diversity – a land where a multitude of cultures, languages, and traditions intricately blend. The "Ek Bharat Shreshtha Bharat" initiative shines as a beacon, illuminating the inherent unity within this diversity and encouraging a deeper connection between our states and regions. To truly appreciate the distinct contributions of each part of our nation to this unified whole, let's embark on a creative project: a scrapbook that showcases the importance of the tourist spots in two unique states, Meghalaya and Uttar Pradesh.

Therefore, this scrapbook, focusing on **"The Importance of Tourist Spots of Meghalaya & Uttar Pradesh,"** will be a testament to the diverse beauty and cultural heritage that strengthens the spirit of "Ek Bharat Shreshtha Bharat."



1. **Earth in Action: A Power Point Presentation Project**



Dive into the world of natural events and environmental changes by choosing one of the following:

- **Earthquakes:** Uncovering the causes and exploring ways to reduce their impact.
- **Floods:** Investigating the reasons behind floods and identifying mitigation strategies.
- **Global Warming:** Examining the factors contributing to global warming and proposing solutions.

Choose the topic that interests you most and share your learning with the class and Develop a visually engaging and informative PowerPoint presentation (6-7 slides) that explains its causes and suggests ways, we can mitigate its effects. Be creative in how you present your information!

2. **Research to Reality: Design Your Own Magazine!** Embark on research journey into one of these important areas:

Consumer Awareness

OR

Social Issues

OR

Sustainable Development

Your goal is to transform your research into a visually appealing and informative magazine. Use a variety of sources to gather information and don't hesitate to incorporate different artistic expressions. Make sure to include relevant visuals to support your content. Your magazine will be evaluated on the depth and accuracy of your research, the creativity of your design, the overall impact of your presentation, and its timely submission.

ARTIFICIAL INTELLIGENCE

1. Mini Project (Practical – Presentation or Report)

Choose **one** of the following topics and make a presentation (PowerPoint/Google Slides) or a report.

NOTE: Submission in soft copy. Email the soft copy at:

holidayhomeworkitai@gmail.com (subject of mail should contain your name, class and section)

- i. **Chatbots:** Create a simple chatbot using Scratch or Python (if you know coding)
- ii. **AI vs Human Intelligence:** Compare how each learns, reacts, and evolves
- iii. **Smart Home Devices:** Research how Alexa, Google Assistant, or Siri use AI

iv. **AI in Games:** Explore how AI is used in games like Chess or FIFA

**2. Make a practical file for the following programs in Python.
Arrange printouts of code and their output in a practical file.**

1. Check if a number is even or odd.
2. Use a for loop to print odd numbers from 1 to 50.
3. Check for a palindrome number.
4. Print the following pyramids, using nested loops.

```
*           1
**          12
***         123
****        1234
```

5. Check if the given number is prime or not.
6. To replace all the list elements with their squares.
7. To calculate and print factorial of a number.
8. To print those list elements which are divisible by 7.

3. Draw the 4W Canvas for the following scenario, in your AI register.

In many schools, managing students' health and ensuring timely medical attention is a growing concern. Teachers often don't notice when a student is feeling unwell unless the student speaks up. Sometimes, students hesitate to report their problems, which can lead to serious health issues going unnoticed. To solve this, your school is planning to introduce an **AI-based Health Monitoring System**. The system will use smart wristbands or fitness trackers to monitor students' basic health indicators like heart rate, body temperature, and physical activity throughout the day. If the system detects any unusual patterns—like high fever, fatigue, or inactivity—it will automatically alert the school nurse or class teacher. This ensures students get quick help and stay healthy, while also reducing the manual work of monitoring large groups of students.

INFORMATION TECHNOLOGY

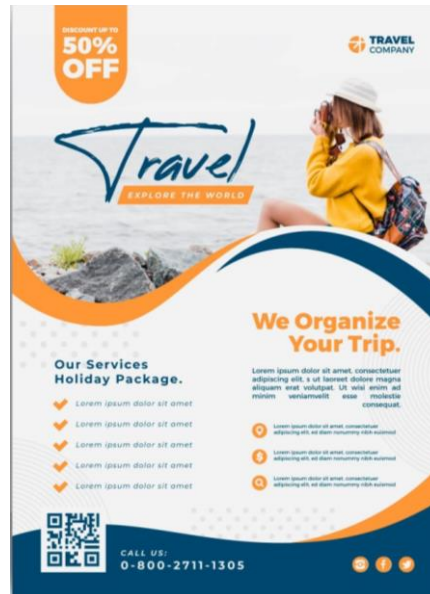
Submit all the holiday homework (digital work) on holidayhomeworkitai@gmail.com

Q1: Design a brochure for a fictional company as per the details given below using Libre office writer/ MS Word/MS Publisher/Canva/ Adobe Express or any other software of your choice. Apply your knowledge of formatting and layout. (An Example of brochure is given below for your reference). Share the image/link after completing the work on the given email id.

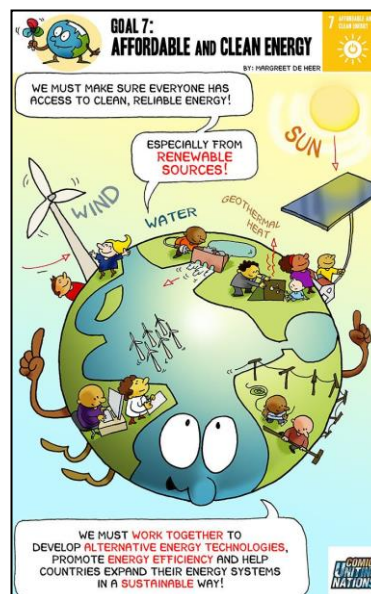
- Name: WanderSphere Travel Co.
- Industry: Travel & Tourism

- Founded: 2010
- Headquarters: Seattle, Washington
- Mission: To craft personalized, immersive travel experiences that leave lasting memories and cultural connections.

WanderSphere specializes in curated, small-group travel packages that focus on authentic local experiences in destinations across Europe, Asia, and South America. Their new initiative, “Journeys That Matter,” emphasizes eco-conscious travel and community-driven tourism.



Q2. Design a poster on any one of the “Sustainable Development Goals” using any software of your choice. Use appropriate quotes/text, graphics, fonts and colour. An example has been shared below:



Q2. Your school is organizing its annual IT event, BOT (Battle of Titans) 2025, aimed at showcasing student innovation, hosting workshops, and inviting guest speakers from the tech industry. As a member of the media and publicity team, you have been assigned the task of creating a promotional video to generate excitement and awareness among students, staff, and potential sponsors. Create a Video for the same using any tool of your choice. Use your creativity to make it appealing. Share the video on mail.

Also, based on this scenario, answer the following questions:

1. What key elements you included in the promotional video to effectively highlight the purpose and attractions of the event?
2. What tools or software you used to create and edit the video, and why?

Q3. On an A4 size sheet, make a table consisting of the list of all the shortcut keys (for Libre office Writer and Calc) given in your textbook, in the first column. Write their functions in the next column. Make sure to design the table using vibrant colours. Learn them and submit the sheet in the school after holidays.

Q4. You are a student of ABC school. You have been asked to help your class teacher prepare the final result sheet for the Final Exam Result by consolidating marks of students from 3 different sections (A, B, C) for three subjects only. Do the following for achieving Data Consolidation for the same.

Steps:

- Create 3 sheets named Section_A, Section_B, and Section_C.
- Each sheet should have: Roll No, Name, and Marks in 3 subjects (Maths, Science, English).
- Each sheet should have a proper heading: School name and session
- Fill in dummy data for 5 students per sheet.
- In a fourth sheet, use Consolidate to combine the marks of all students from all sections into a single master sheet.
- Use the "Sum" function while consolidating.
- Save your file as: Consolidated_Marks_YourName.ods

HINDI

मयूर स्कूल, नोएडा
ग्रीष्मकालीनावकाश गृहकार्य
कक्षा-दसवीं
हिंदी

“हर दिन कुछ नया सीखो,
क्योंकि ज्ञान ही सबसे बड़ी
ताकत है। 📖💡



प्रश्न- 1 पाठ 'हरिहर काका' के आधार पर बताइए कि भारत में वरिष्ठ नागरिकों के स्वास्थ्य, सुरक्षा, सामाजिक सुरक्षा तथा कल्याण को बढ़ावा देने के लिए केंद्र सरकार 1999 में बुजुर्गों के लिए जो राष्ट्रीय नीति लेकर आई, उसी तरह किन्हीं अन्य तीन प्रावधानों का उल्लेख करते हुए परियोजना फाइल बनाएँ।

अथवा

आपकी पाठ्य पुस्तक संचयन में पाठ 'हरिहर काका' को पूरा पढ़ें। उसके पश्चात अपने माता-पिता के समय की जीवन शैली यथार्थ जीवन जीने का ढंग और विचारों तथा वर्तमान समय में आपके जीवन जीने का ढंग और विचारों का तुलनात्मक अध्ययन करें। इस अध्ययन में निम्न विषयों को समाहित करें-

- आपसी रिश्तों में मेल मिलाप और प्रेम
- पौष्टिक भोजन का स्तर
- जीवन में तनाव
- संयुक्त परिवार, पड़ोस व बड़े बुजुर्गों के प्रति सोच
- शिक्षा प्रणाली और नैतिक मूल्य

सहायक बिंदु-

- ✓ विद्यार्थियों द्वारा सरकारी योजनाओं को पढ़ना
- ✓ भावों की निरंतरता
- ✓ शिक्षा

मूल्यांकन बिंदु-

- ❖ विषय-वस्तु -3
- ❖ प्रस्तुती -1
- ❖ भाषा -1



प्रश्न-2 पाठ 'डायरी का एक पन्ना' के आधार पर यह स्पष्ट होता है कि स्वतंत्रता आंदोलन में महिलाओं ने भी बढ़-चढ़ कर हिस्सा लिया स्वतंत्रता आंदोलन में महिलाओं ने जो योगदान दिया, उसके बारे में संक्षिप्त जानकारी प्राप्त करके लिखिए ।

सहायक बिंदु- ✓ सामाजिक पृष्ठभूमि ✓ महिलाओं का योगदान ✓ शिक्षा	मूल्यांकन बिंदु- ❖ विषय-वस्तु -3 ❖ प्रस्तुती -1 ❖ भाषा -1	
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प्रश्न- 3 'डिजिटल इंडिया' विषय पर सरकार की ओर से जनहित में जारी करते हुए एक आकर्षक विज्ञापन तैयार कीजिए ।

सहायक बिंदु- ✓ आकर्षक चित्र ✓ उपयोगिता ✓ शिक्षा	मूल्यांकन बिंदु- ❖ रचनात्मकता -3 ❖ प्रस्तुती -1 ❖ भाषा -1		
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प्रश्न- 4 'भाग्य और पुरुषार्थ' विषय से संबंधित सचित्र लघु कथा लिखिए ।

अथवा

'एक आस-एक प्रयास' शीर्षक: 'एक आस ने बदली जिंदगी' विषय से संबंधित सचित्र लघु कथा लिखिए ।

सहायक बिंदु- ✓ परिभाषा ✓ भाग्य व पुरुषार्थ की भूमिका ✓ शिक्षा	मूल्यांकन बिंदु- ❖ विषयवस्तु -3 ❖ प्रस्तुती -1 ❖ भाषा -1		
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- सभी कार्य ए-4 आकार के पृष्ठ पर कीजिए।
- ब्लॉक टेस्ट के लिए संपूर्ण पाठ्यक्रम का अभ्यास कीजिए।

GERMAN

ADJEKTIVENDUNGEN--ÜBUNGEN

1. Das ist der krank__Mann, die krank____Frau, das krank____Kind. Das sind die krank__Menschen.
2. Ich esse den alt__Apfel, die alt____Birne und die alt____Bananen.
3. Wir helfen dem alt__Mann, der alt____Frau und den alt__Leute__.
4. Ich esse gern frisch__Brot.
5. Ich möchte ein frisch__Brot.
6. Ich möchte das frisch__Brot.
7. Das Brot ist frisch__.
8. Wir essen ein lecker__Butterbrot mit frisch____Schinken , frisch____Wurst und viel____Tomaten.
9. Barney isst jeden Tag ein roh__Ei, eine klein____Katze und einen klein____Hund.

Setzen Sie die Endungen ein! Szene I: Vor einem klein__Garten

An einem hell____Morgen trifft der hübsch____Romeo die schön__Julia auf der eng____Straße vor dem klein____Garten. Romeo sagt: "Lieb____Julia, deine neu____Jeans gefallen mir und deine gelb__Jacke ist wunderbar__."---"Ja, lieb____Romeo, ich habe sie aus der klein__Boutique."---"Oh, süß____Julia, deine grün____Augen sind wunderschön____." Der begeistert__Romeo springt von sein____weiß____Pferd und rennt gegen ein____groß____Tannenbaum. "Oh, arm____Romeo, deine alt____Brille ist kaputt____." Romeo geht weiter und fällt in schmutzig__Wasser. "Oh, arm____, nass____Romeo, geh zum groß____Optiker und kauf dir neu____Kontaktlinsen!"

Am Abend vor dem groß____Haus der schön____Julia

Romeo kommt mit ein____rot____Rose. "Nanu, hier sind ja zwei klein____Balkons.

Welcher geht in Julias groß____Zimmer? Leider hat der nett____Optiker meine

alt____Brille noch nicht repariert." Romeo nimmt seine klein____Gitarre, spielt eine schön____

____Melodie und singt ein klein____Liebeslied. Julia kommt aus der geöffnet____Tür ihr____[=of her] klein____Balkons. Ihr wütend__Vater steht hinter d__schön____Blumen von d____ander____, link____Balkon.

Romeo sieht niemanden und geht leider zur link__Seite, wo der böse____Vater ist. Oh weh, arm____Romeo, oh weh!

ÜBUNG B

Beschreiben Sie Ihr Zimmer mit vielen Adjektiven

Poster – Meine Traum-Schulkantine

Task:

Design a colourful and informative poster in German with the title: „Meine Traum-Schulkantine“

You must include:

- A creative name for your canteen (z. B. „Fit & Frisch Kantine“)
- A weekly menu from Montag bis Freitag (Include Frühstück, Mittagessen, Snacks)
- Draw or paste pictures of 3–4 favourite dishes
- Label each dish in German
- Imaginary student reviews in German (2–3 sentences)

Use phrases like:

„Das Essen ist lecker.“

„Ich finde den Salat sehr frisch.“

„Mir schmeckt der Kartoffelauflauf besonders gut.“

Things to Remember:

- Write in clear, legible German
- Make your poster bright, clean, and creative
- Use borders, fonts, and drawings

Presentation Format:

A3 sized sheet or paper

FRENCH

1. Ecrivez une message à votre ami/amie pour l'inviter à la soirée que vous organisez pour fêter l'anniversaire de votre soeur/frère.
2. Individual Activity - Conduct a comparative research study on the education systems of France and India. Highlight key similarities and differences, and present your findings creatively in an elaborated project. (3-4 A4 size sheets)
3. Lisez une histoire ce que vous aimez, et ecrivez un paragraphe en bref.(Read any one stories and write a brief paragraph (max 80 – 90 words) in French describing the stories.) <https://www.thefrenchexperiment.com/stories>

SANSKRIT

1-बिहारराज्योत्पत्ति, भौगोलिक स्थिति; भाषा-संस्कृति; कला-साहित्य; दर्शनीय स्थलानां वर्णनं च अस्मिन् विषये संस्कृत भाषायां लेखित्वा एकं चलचित्रं प्रस्तुतं कुरुत्- (पी.पी.टी.मध्यमेन)



मूल्यांकन बिंदु- रचनात्मकता-3
प्रस्तुतिकरण-1
विषय उपयोगिता-1

2-कन्यां रक्षतु कन्यां पाठयतु" अस्मिन् विषये संस्कृतभाषायां एकं चलितचित्रं प्रस्तुतं कुरुत् (पी.पी.टी.माध्यमेन)



मूल्यांकन बिंदु- रचनात्मकता-3
प्रस्तुतिकरण-1
विषय उपयोगिता-1

3-सरस्वती व्याकरण पुस्तक से औपचारिक पत्र का अभ्यास करते हुए 4 पत्र लिखें।

मूल्यांकन बिंदु- रचनात्मकता-3
प्रस्तुतिकरण-1
विषय उपयोगिता-1